

Full Length Article

Surface modification of nanotwinned copper and SiCN using N₂ and Ar plasma activationRou-Jun Lee^a, Pin-Syuan He^b, Dinh-Phuc Tran^b, Wei-Lan Chiu^c, Hsiang-Hung Chang^c, Chang-Chun Lee^d, Chih Chen^{b,*}^a Industry Academia Innovation School, National Yang Ming Chiao Tung University, Hsinchu 300093, Taiwan^b Department of Materials Science and Engineering, National Yang Ming Chiao Tung University, Hsinchu 300093, Taiwan^c Electronic and Optoelectronic System Research Laboratories, Industrial Technology Research Institute (ITRI), Hsinchu 310401, Taiwan^d Department of Power Mechanical Engineering, National Tsing Hua University, Hsinchu 300044, Taiwan

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ABSTRACT

Nanotwinned copper (NT-Cu) and silicon carbon nitride (SiCN) films were modified using N₂/Ar plasmas to enhance their bonding strength. X-ray photoelectron spectroscopy (XPS), atomic force microscopy (AFM), and contact angle analyses were then conducted. The contact angles of the NT-Cu films significantly reduced, dropping from 54.8° to 32.1°, while that of the SiCN sharply declined from 51.1° to under 10°. A significant increase in the Si-O bonds was found in the SiCN after the plasma treatment. These factors thus contributed to the enhanced bonding strength (up to 30 and 27.4 MPa for the NT-Cu/NT-Cu and SiCN/SiCN, respectively). Additionally, we performed the SiCN/SiCN room temperature bonding and evaluated its bonding energy using four-point bending tests. We found that the bonding energy increased from 0.49 to 1.2 J/m² after annealing at 100 °C for 30 min. Transmission electron microscopy (TEM) image revealed some nanovoids at the bonding interface as bonded at room temperature. These voids could be eliminated by further annealing at 250 °C for 1 h. This study offers insights into the plasma surface modification to enhance the bonding strength of NT-Cu/NT-Cu and SiCN/SiCN.

1. Introduction

The increasing demand for high I/O density driven by the rapid development in artificial intelligence and 5G internet [1,2] requires significant scaling of interconnect pitch. When the pitch shrinks to below 10 μm, conventional solder interconnects face many challenges. Three-dimensional integrated circuit (3D IC) packaging is expected to solve these issues, which can be achieved by hybrid bonding [3–5]. Hybrid bonding involves the direct bonding of dielectric pads followed by thermal compression bonding of Cu vias. Different Cu materials can be used such as conventional Cu, nanotwinned copper (NT-Cu), and nanocrystalline Cu [6,7]. Common dielectric materials for hybrid bonding include SiO₂, SiN, and SiCN [8]. Currently, Cu/SiO₂ hybrid bonding is the most widely adopted and well-researched technology [9–11]. As the pitch continues to shrink, misalignment becomes more likely, reducing the lifetime of dielectric joints [12]. Therefore, a dielectric material that can suppress Cu diffusion while being compatible with hybrid bonding is needed.

In recent years, various companies (IMEC and Samsung) have focused on applying plasma for Cu/SiCN hybrid bonding [13–16]. SiCN is an alternative to SiO₂ for several reasons, including its superior capability as a Cu diffusion barrier [17] and its relatively high bonding energy at low temperatures [18]. This not only helps reduce the thermal budget but also ensures that the dielectric bond can withstand the stress caused by Cu expansion during annealing. The bonding interface of SiCN/SiCN contains fewer voids compared to SiO₂/SiO₂ [19]. Under identical plasma conditions, SiCN surface forms more dangling bonds than SiO₂ [20]. These dangling bonds can react with residual water at the bonding interface to form additional silanol bonds, thereby preventing the formation of nanovoids [21]. Additionally, owing to the high surface diffusivity [22], low oxidation rate [23], and high strength [24], NT-Cu is beneficial for advanced packaging. The bonding temperature can be lowered to 200 °C while maintaining its high bonding strength [25].

As aforementioned, Cu/SiCN hybrid bonding requires plasma to activate SiCN surfaces. However, Cu surfaces are also exposed to plasma.

* Corresponding author.

E-mail address: chihchen@nycu.edu.tw (C. Chen).<https://doi.org/10.1016/j.apsusc.2024.161832>

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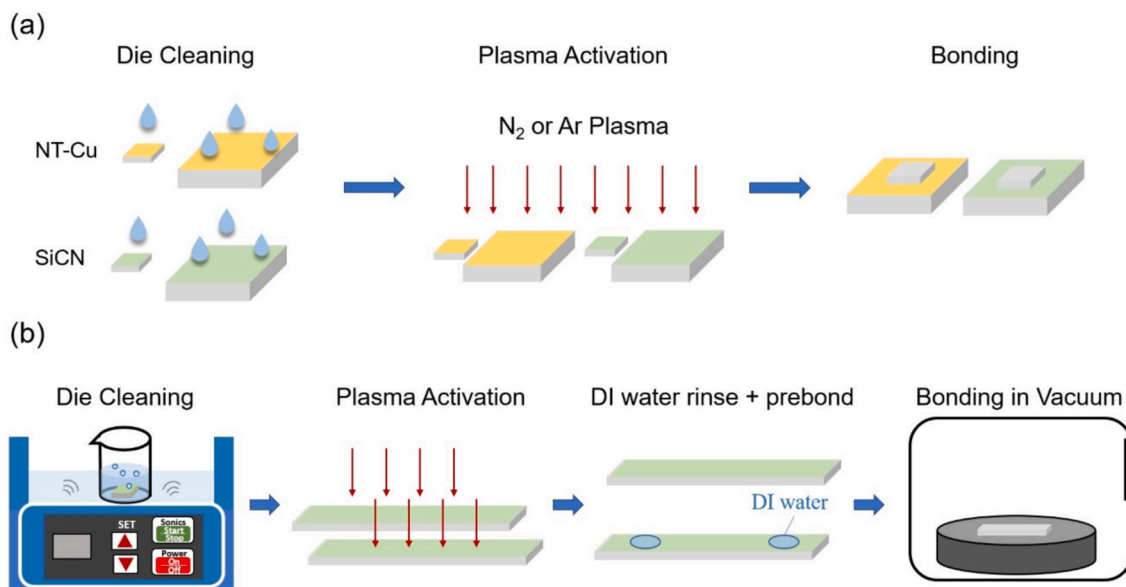


Fig. 1. Schematic process flow of the (a) NT-Cu/NT-Cu and SiCN/SiCN thermal compression bonding and (b) SiCN/SiCN room temperature bonding procedure.

The ion bombardment can increase the surface roughness, resulting in the formation of copper oxides and potentially reducing the bonding quality. Therefore, it is essential to study the surface characteristics of NT-Cu/SiCN after plasma treatment. The effects of N₂ and Ar plasma were then studied using X-ray photoelectron spectroscopy (XPS), atomic force microscopy (AFM), and contact angle measurements. The bonding strength of NT-Cu/NT-Cu and SiCN/SiCN joints was evaluated using shear tests. Additionally, SiCN/SiCN bonding was conducted at room temperature without bonding pressure, and the bonding energy was further measured by four-point bending tests.

2. Experimental

2.1. Preparation of < 111 > oriented NT-Cu and SiCN dies

Highly < 111 >-oriented NT-Cu die with a thickness of 6 μm was fabricated by DC electroplating. Simultaneously, a 180 nm-thick SiCN die was deposited by plasma-enhanced chemical vapor deposition (PECVD). The SiCN composition consists of Si (46.3 at.%), C (25.6 at.%), N (25.5 at.%), O (2.6 at.%). Both the NT-Cu and SiCN dies were subjected to chemical mechanical polishing (CMP) to reduce their surface roughness. Subsequently, the samples were diced into sizes of 6 $\times$ 6 mm² (top dies) and 15 $\times$ 15 mm² (bottom dies). Prior to plasma activation, NT-Cu pieces were cleaned with acetone, isopropyl alcohol, citric acid, and DI water to eliminate the organic contamination and surface oxides. SiCN samples were rinsed with a TEAH/H₂O₂ solution for surface cleaning, followed by a DI water rinse. The study can be divided into two parts. The first part discusses the thermal compression bonding of NT-Cu/NT-Cu and SiCN/SiCN, while the second part focuses on the room temperature bonding of SiCN/SiCN. The experimental procedure is illustrated in Fig. 1.

2.2. Plasma activation and bonding

In hybrid bonding, the dielectrics are initially pre-bonded at room temperature, followed by an annealing process that enables the recessed Cu pads to expand and ultimately form Cu-Cu bonds. Throughout this process, the bonding strength of the dielectric bonds must be high enough to withstand the induced stress due to the expansion of Cu pads. Therefore, we analyze the two bonding processes separately to characterize the surfaces of NT-Cu and SiCN, aiming to identify the optimal bonding conditions.

In the first part of the study, we examined the influence of plasma activation on NT-Cu and SiCN surfaces and evaluated the bonding strength of NT-Cu/NT-Cu and SiCN/SiCN dies. In brief, N₂ and Ar plasma were used at power levels of 15 W, 20 W, and 30 W, with a gas flow of 20 sccm for 2 min. After the surface activation, NT-Cu/NT-Cu and SiCN/SiCN dies were bonded at 200 $^{\circ}\text{C}$ and 150 $^{\circ}\text{C}$, respectively. The bonding time was fixed at 30 min under a pressure of 33 MPa using a bonder (WB-L3000). For comparison, NT-Cu and SiCN without plasma treatment were also prepared.

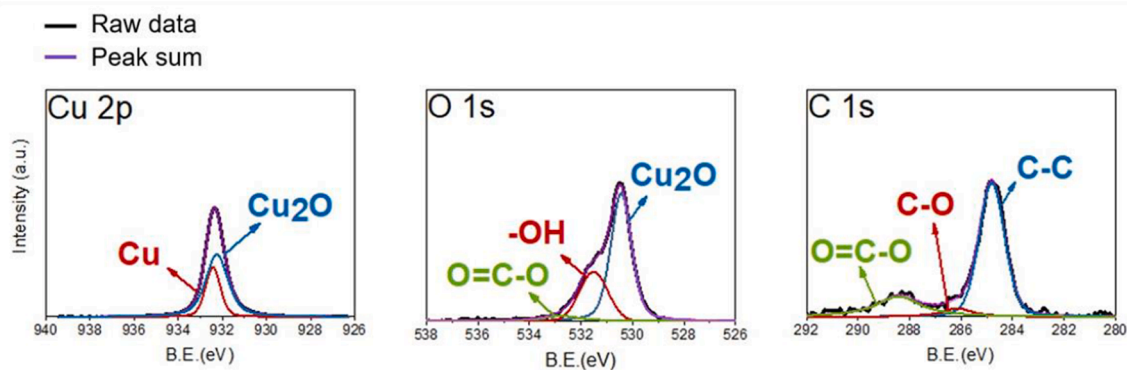
In the second part of this study, we bonded SiCN/SiCN at room temperature. SiCN samples were diced into 5 cm $\times$ 1 cm prior to four-point bending tests. For the surface activation, we sequentially employed Ar and N₂ plasmas. Ar plasma was used for 1 min for surface cleaning, followed by N₂ plasma for 2 min for surface activation. Afterwards, two drops of 1.3 μL DI water were placed at the dies and then pre-bonded at room temperature. Rinsing the SiCN surface with DI water allows the water molecules to react with dangling bonds, forming silanol bonds (Si-OH), which can then undergo a polymerization reaction to create siloxane bonds (Si-O-Si). This process may not be needed when SiCN/SiCN is bonded under high pressure and annealing temperatures. However, DI water rinsing is crucial for room temperature bonding, as it ensures the surface has enough silanol bonds to form sufficient siloxane bonds, providing adequate bonding strength. To complete the bonding process, the bonded pairs were placed in a vacuum chamber for 5 min. For comparison, additional experiments were conducted. After placing them in a vacuum chamber, the bonded pairs were annealed at varying temperatures and durations.

2.3. Characterization

The changes in surface chemical composition of NT-Cu and SiCN before and after the plasma treatment were analyzed using XPS (Thermo Fisher Scientific ESCALAB Xi+, Waltham, MA, USA) with a micro-focused monochromatic Al K α (h ν = 1486.6 eV) source. The surface roughness was evaluated by AFM (Veeco D3100, New York, USA) with a scan area of 5 $\times$ 10 μm^2 . The hydrophilicity of sample surfaces was determined by a contact angle meter (FTA-125, First Ten Angstroms, Newark, CA, USA) using a 0.1 μL droplet of DI water in ambient.

To examine the bonding interface, transmission electron microscope (TEM, JEM-F200, JEOL, Akishima, Tokyo, Japan) and focused ion beam (FIB, GAIA 3, TESCAN, Brno, Czechia) were used. The atomic composition of the SiCN bonding interface was investigated by energy

(a)



(b)

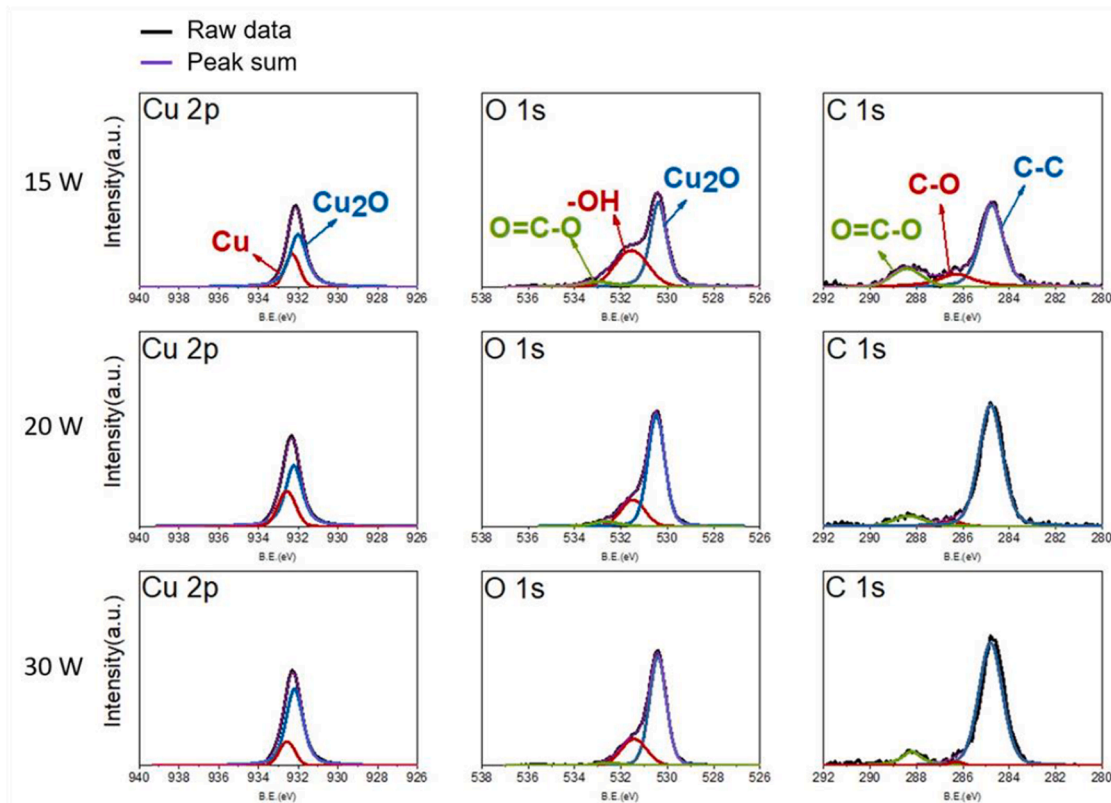


Fig. 2. XPS spectra of the NT-Cu: (a) without, treated by (b) N₂ and (c) Ar plasma with different plasma powers of 15 W, 20 W, and 30 W.

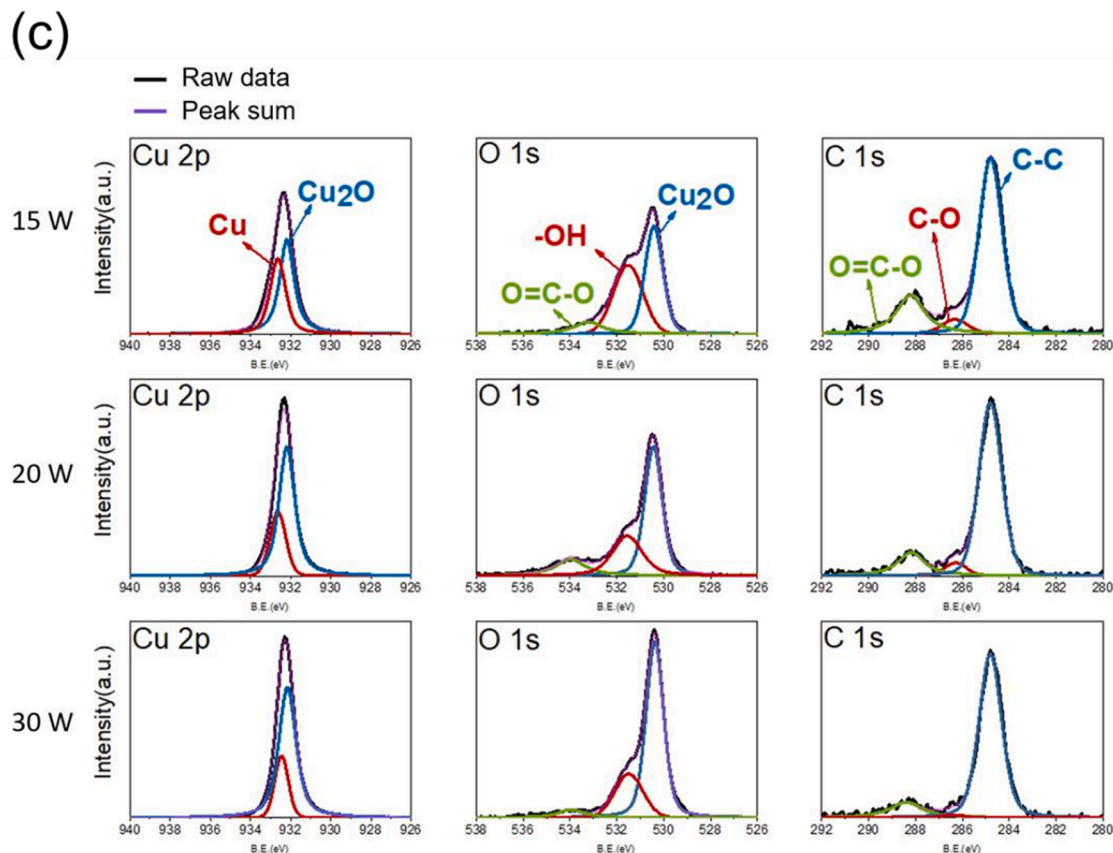


Fig. 2. (continued).

dispersive X-ray spectroscopy (EDS). Scanning acoustic tomography (SAT, CSAM, Nordson SONOSCAN-Gen6, Elk Village, IL, USA) was adopted to calculate the bonding ratio. Finally, shear tests with a shear height of 200 μm and a shear speed of 100 $\mu\text{m/s}$ were conducted to measure the bonding strength of NT-Cu/NT-Cu and SiCN/SiCN samples.

3. Results and discussion

3.1. NT-Cu/NT-Cu and SiCN/SiCN thermal compression bonding

To examine the changes in chemical bonds at the surfaces of NT-Cu before and after plasma activation, XPS analysis was conducted. The XPS spectra of NT-Cu before and after N_2/Ar plasma activation are shown in Fig. 2. Cu ($\text{Cu}2\text{p}_{3/2}$), O (1s), and C (1s) were detected in all samples. In the Cu 2p spectra, the binding energy at 932.63 eV and 932.18 eV could be attributed to Cu and Cu_2O , respectively [26,27]. The presence of Cu_2O was further confirmed by a prominent peak at 530.4 eV in the O1s spectra [27]. No satellite peak of CuO and $\text{Cu}(\text{OH})_2$ was found. This finding indicated that the NT-Cu surface mainly contained Cu and cuprous oxide. The formation of Cu_2O on the NT-Cu surface was due to the oxidation reaction of Cu and oxygen in the ambient environment. In the O 1s spectra, the weak peak at 533 eV was assigned to $\text{O}-\text{C}=\text{O}$ functional group, which was attributed to citric acid residues on the NT-Cu surface. This $\text{O}-\text{C}=\text{O}$ group was also identified in the C 1s spectra.

As the plasma power was increased, the intensity of Cu_2O also increased. Regardless of whether N_2 or Ar plasma was used, the NT-Cu surfaces consisted of Cu_2O . Although the surface contained cuprous oxide, this did not affect the bonding of NT-Cu/NT-Cu. As mentioned earlier, we used the low plasma power to prevent serious Cu oxidation. In 2023 ECTC, Netzband et al. calculated that the diffusion length of Cu at Cu_2O during a 30-min annealing at 200 $^\circ\text{C}$ is 8.45 nm [28]. In this plasma treatment, cuprous oxide was present only in the topmost layer,

through which Cu atoms could easily diffuse during annealing. In the die-to-wafer bonding process, the dies first undergo plasma treatment to activate the dielectric surfaces, which are then brought out of the vacuum for pre-bonding. The N_2/Ar plasma makes the Cu surface highly reactive. The Cu surface may oxidize outside vacuum chambers, resulting in the formation of cuprous oxide. Hence, we did not clean the Cu surfaces after plasma activation to stimulate the actual die-to-wafer bonding process. The Cu surfaces were only cleaned prior to the plasma activation process.

The bonding interfaces were then examined using FIB-SEM, TEM, and EDS. Fig. 3(a, c) illustrates the NT-Cu/NT-Cu bonding interface treated with N_2 plasma at 20 W, whereas Fig. 3(b, d) depicts the interface treated with Ar plasma. Both samples were bonded at 200 $^\circ\text{C}$ for 30 min under a bonding pressure of 33 MPa. From the FIB-SEM images, no apparent voids were found at these bonding interfaces. We then used TEM for a higher resolution observation. As shown in Fig. 3(c, d), after being exposed to either N_2 or Ar plasma, NT-Cu and NT-Cu were successfully bonded. Moreover, the EDS profiles (Fig. 3(e, f)) show no traces of O at either bonding interfaces.

The chemical bonds at the SiCN before and after plasma activation were also analyzed by XPS. Fig. 4(a) shows the XPS spectra of the SiCN without plasma treatment, whereas Fig. 4(b) and Fig. 4(c) display the XPS results of the SiCN subjected to N_2 and Ar plasma treatments, respectively. As shown in Fig. 4, the Si-N, Si-C, and Si-O peaks located at 101.5, 100.8, and 103 eV in the Si 2p spectra [29]. Fig. 4 shows that the Si atoms in the SiCN film mainly bonded to N atoms, with some partial bonds to C and O atoms. The Si-O bonds resulted from the formation of a native oxide layer on the surface. The N-C peak at 398.3 eV [30] in the N 1s spectra shows that some N atoms bonded to C atoms. In Fig. 4, the intensity of the Si-O bonds gradually increased as the plasma power increased and eventually became compatible with the Si-N bonds. Besides, a gradual decrease in the intensity of the Si-N bond was observed.

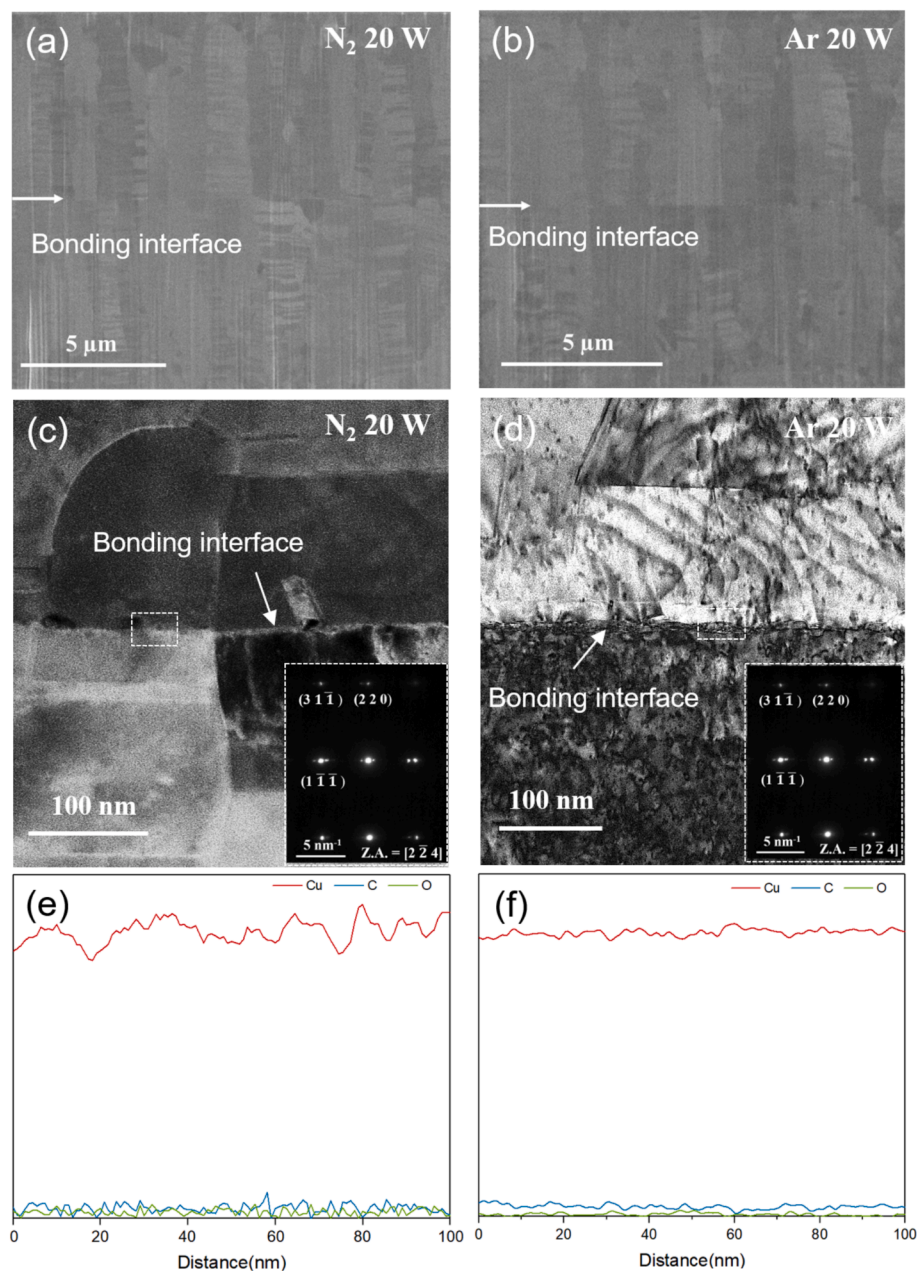


Fig. 3. Bonding interfaces of the NT-Cu/NT-Cu joints treated with (a,c) N_2 and (b,d) Ar plasma at 20 W for 2 min. The white arrows indicate the bonding interfaces. The (a,b) FIB-SEM and (c,d) TEM images show that the NT-Cu/NT-Cu joints were successfully bonded after both plasma treatments. The diffraction patterns taken at the bonding interfaces are shown in the window figures in (c,d). No obvious traces of O at the bonding interfaces were detected by EDS analysis (e,f).

This proves that both plasma treatments broke the Si-N bonds. However, the intensity of N-C bonds in the N 1s and C 1s spectra remained relatively low, indicating that it was not affected in this plasma treatment. Moreover, from the C 1s spectra, we observed a gradual decline at 283.4 eV, corresponding to the C-Si bond, as the plasma power increased. This implies that the plasma treatment also broke Si-C bonds. Thus, we can conclude that Si-N and Si-C bonds were broken by the plasma activation, and a considerable portion of Si-O bonds were formed. This conclusion is further supported by the TEM and EDS analysis. Fig. 5 shows the SiCN/SiCN bonding interfaces treated with N_2 and Ar plasma at 20 W prior to bonding at 150 °C for 30 min. The EDS line scan reveals a sharp rise in O content at the bonding interface, followed by the decreases in Si, C, and N contents. Hence, it can be concluded that the bonding interface contained an oxide layer. The thicknesses of these oxide layers were 2.87 nm and 4.87 nm. Besides, after calibrating the XPS data using the atomic

sensitivity factor, we noticed that the O contents after being treated by either N_2 or Ar plasma at 15 W, 20 W, and 30 W were about 22%, 35%, and 40%, respectively. This shows that the degree of oxidation is dependent on the plasma power rather than the type of plasma.

AFM was also used to analyze the impact of plasma activation on surface roughness since ion bombardment could damage the surface. Fig. 6 illustrates the changes in the surface roughness of NT-Cu and SiCN before and after plasma activation. The roughness (R_q) of the initial NT-Cu and SiCN was 1.67 nm and 0.19 nm, respectively. The roughness of NT-Cu increased after the N_2 plasma treatment, but it was still lower than 4 nm. As for the SiCN, the roughness stayed below 0.2 nm [18], which is crucial for dielectric bonding as high surface roughness means less contact area, leading to weaker bonding strength. The AFM results showed that the plasma treatment did not significantly damage the surface roughness of NT-Cu and SiCN.

Additionally, the contact angle measurements were conducted to analyze the effect of plasma on the surface hydrophilicity of the NT-Cu and SiCN. Fig. 7 shows the contact angles of NT-Cu and SiCN under various plasma conditions. The contact angle of the initial NT-Cu was 54.8° . It significantly decreased after the activation process. This result aligns with the previous findings of Park et al. [31], where an increase in the plasma power lowered the contact angle, as ion bombardment slightly altered the NT-Cu surface to a more porous state. The initial contact angle of SiCN was 51.1° , and it significantly dropped after the plasma activation, falling below ten degrees at 20 W and 30 W. Fig. 8 shows the contact angle of SiCN before and after plasma activation. After plasma exposure, the contact angle dramatically decreased, causing water to spread evenly across the SiCN surface. The XPS and contact angle results suggested that the SiCN surfaces became highly hydrophilic after plasma treatment due to the formation of Si-O bonds. A hydrophilic surface is prone to water adsorption, leading to the formation of more silanol bonds (Si-OH), which is advantageous for forming siloxane bonds (Si-O-Si). Furthermore, the increase in Si-OH bonds helps bring the bonding interface surfaces closer together through the attraction of hydrogen bonds. The correlation of contact angle and storage time was also examined, as shown in Fig. 8(c). The contact angle of SiCN after N_2 or Ar plasma treatment was approximately 10° . It increased to 15° and 23° after 3 and 5 h storages in the ambient, respectively.

SAT was also used to examine the bonding area prior to the shear tests. Fig. 9 is the SAT images of the NT-Cu/NT-Cu and SiCN/SiCN joints,

showing that most areas were well bonded. The unbonded area, as shown in the red dashed line, was excluded when calculating bonding strength. The bonding strength of NT-Cu/NT-Cu treated with various plasma powers is shown in Fig. 10(a). All NT-Cu films were bonded at 200°C for 30 min. The shear strength of NT-Cu/NT-Cu increased from 15.58 MPa to approximately 30 MPa. Previously, Ar/ N_2 plasmas were also used to assist the Cu-Cu bonding [32,33]. They achieved shear strengths ranging from 15 to 25 MPa. A Cu_4N passivation layer was induced to suppress the Cu surface oxidation but required a high annealing temperature (300°C) for the decomposition of Cu_4N . In contrast, the N_2 plasma power used in this work does not form Cu_4N . By using the NT-Cu, the bonding temperature was lowered to 200°C while achieving even higher shear strength. Fig. 10(b) shows the bonding strength of SiCN/SiCN treated by various plasma powers. The SiCN films were bonded at 150°C for 30 min. The shear strength of the SiCN bonding without plasma treatment was so weak. As shown in Fig. 4, the SiCN without plasma treatment did not have enough Si-O bonds. Thus, water could diffuse into the bonding interface, breaking the siloxane bonds into two silanol bonds, a process known as water stress corrosion [34]. Nevertheless, the shear strength of SiCN/SiCN after plasma activation could reach 27.4 MPa, which was comparable to that of the NT-Cu.

3.2. SiCN/SiCN room temperature bonding

In this part, the SiCN/SiCN bonding was conducted at room tem-

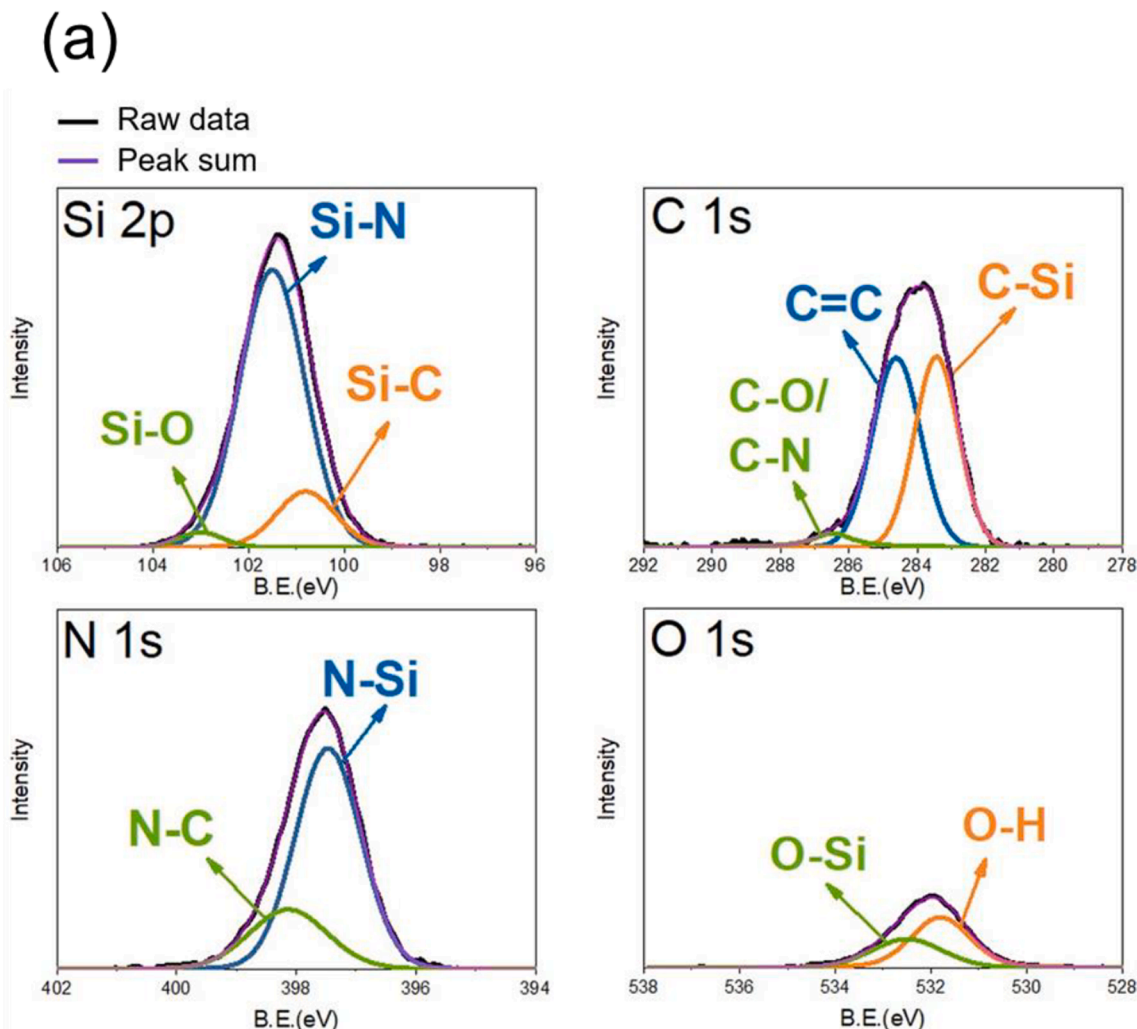


Fig. 4. XPS spectra of the SiCN (a) without, with (b) N_2 plasma and (c) Ar plasma treatments by different plasma powers.

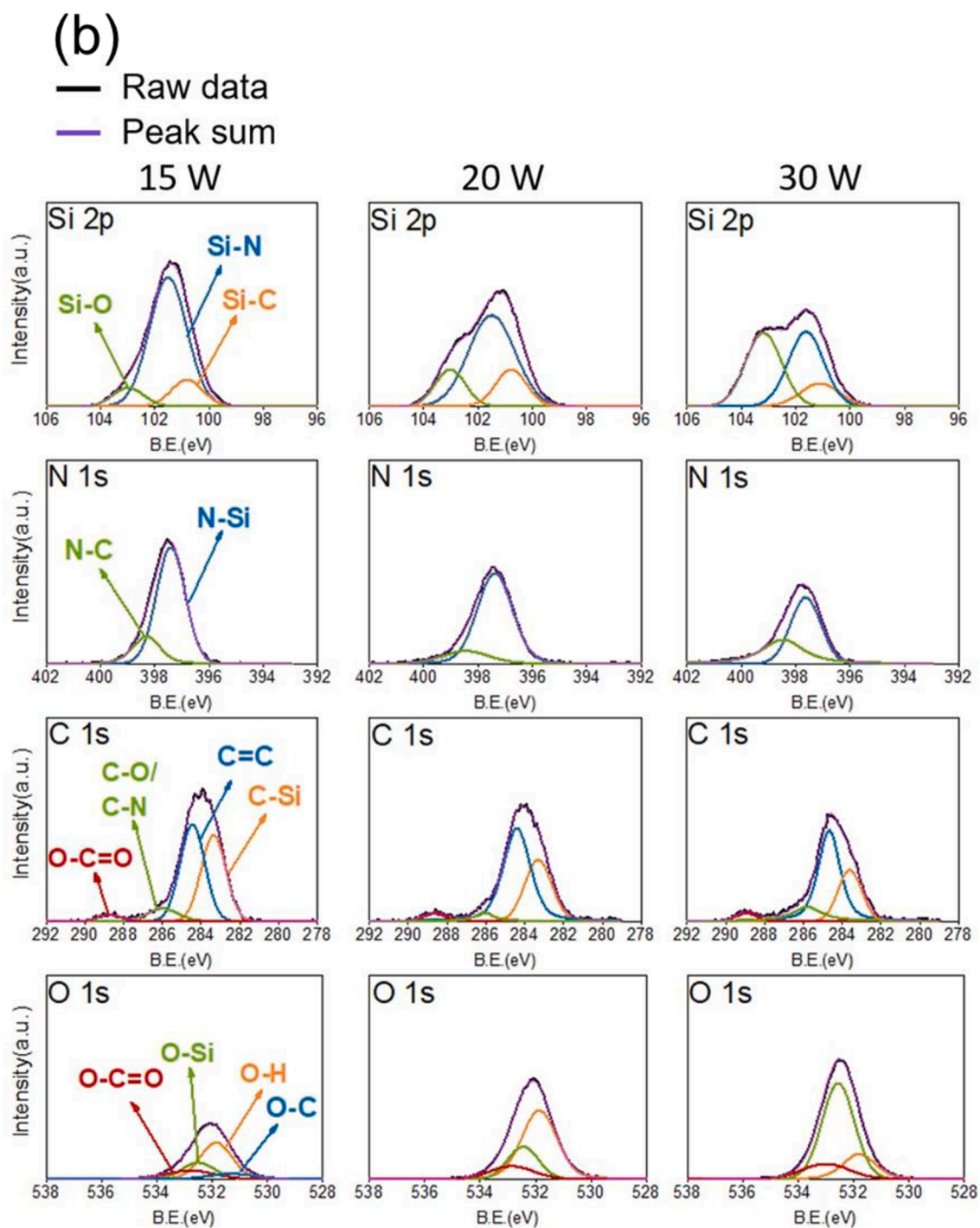


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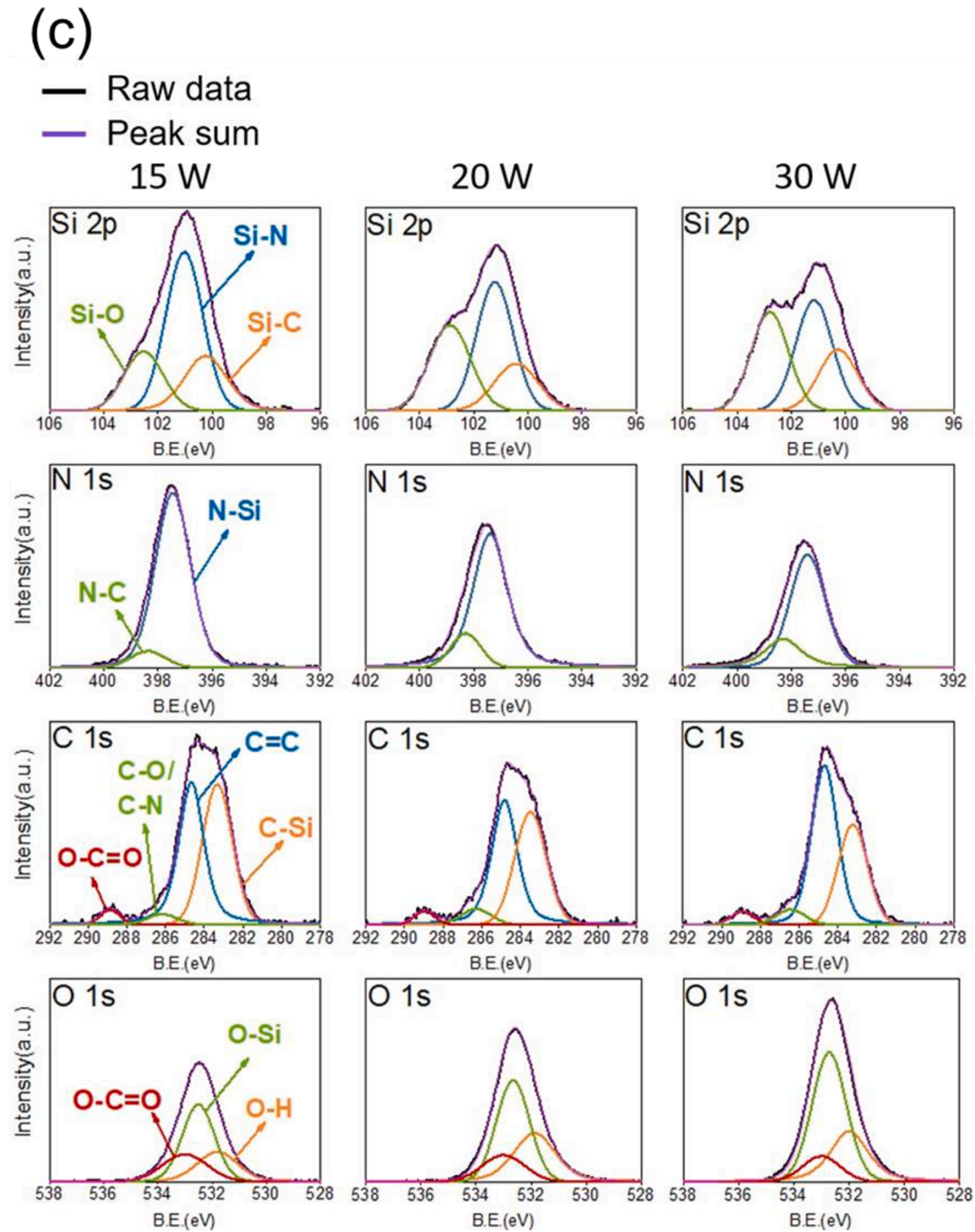


Fig. 4. (continued).

perature without bonding pressure. The bonding energy was then measured by four-point bending tests. Fig. 11(a) shows the illustration of the four-point bending measurements. The pin spacing where the force applied is 2 cm, and the bottom pin spacing is 4 cm. A notch was fabricated to a depth equivalent to 90 % of the silicon thickness in the top die. The tests were conducted at a rate of 0.05 $\mu\text{m/s}$. We can obtain a typical load–displacement curve, as shown in Fig. 11(b). Point A is the moment when the notch cracks thoroughly, resulting in a sudden load drop. The crack then propagates to the weakest spot, which is the SiCN/SiCN bonding interface. The bonding interface starts to delaminate, with a steady crack growth at the interface, as shown in the plateau area B. The bonding energy G (J/m^2) can be calculated by Eq. (1) [35],

$$G\left(\frac{\text{J}}{\text{m}^2}\right) = \frac{21(1-\nu^2)M^2}{4Eh^3} = \frac{21(1-\nu^2)}{4Eh^3} \left[\left(\frac{p}{2}\right) \left(\frac{l}{b}\right) \right]^2$$

where p is the average load calculated from the plateau area B, E is Si Young's Modulus ($E = 174$ GPa), ν is Si Poisson's ratio ($\nu = 0.27$), h is specimen thickness ($h = 775 \times 10^{-6}$ m), l is the pin spacing ($l = 0.01$ m), b is the specimen width ($b = 0.01$ m). Fig. 12 shows the bonding energies of the SiCN/SiCN bonded at room temperature and after post annealing at 100 $^\circ\text{C}$ for 30 min. Investigation on the bonding energy of SiCN bonded at room temperature and without annealing is limited. In this work, we successfully measured the bonding energy of SiCN/SiCN

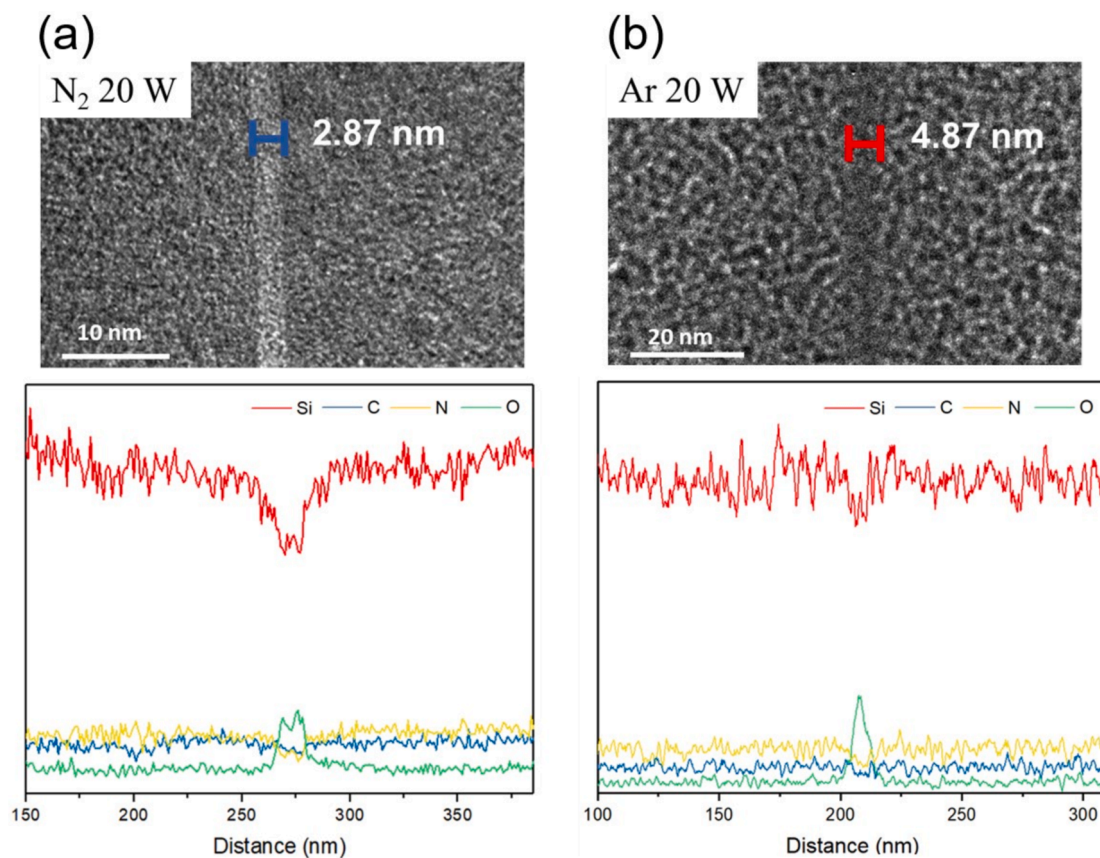


Fig. 5. TEM images and EDS profiles of the SiCN/SiCN treated with (a) N_2 and (b) Ar plasma then bonded at 150 °C for 30 min under 33 MPa.

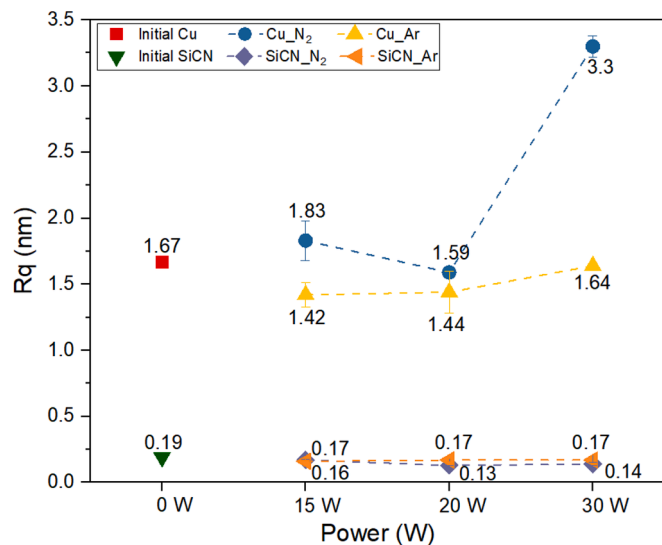


Fig. 6. AFM results of the NT-Cu and SiCN before and after all plasma treatments. Before plasma activation, the roughness of NT-Cu and SiCN was 1.67 and 0.19 nm, respectively. After plasma treatment, the roughness of NT-Cu remained below 4 nm, whereas that of SiCN was below 0.5 nm.

bonded at room temperature to be 0.49 J/m². It increased to 1.2 J/m² after annealing at 100 °C for 30 min. This was due to the post-annealing step, which facilitated the formation of Si-O-Si bonds and also strengthened the covalent bonds [36].

The bonding interface of SiCN/SiCN was also examined using TEM and EDS. Fig. 13 shows the TEM images of the bonding interface of

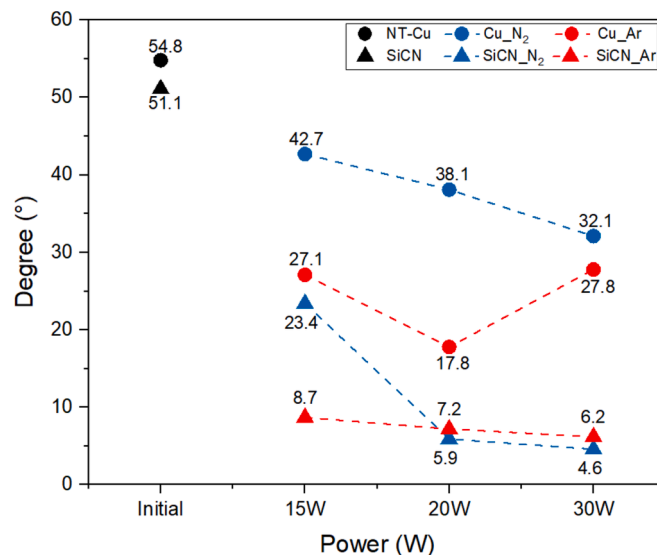


Fig. 7. Contact angles of the NT-Cu and SiCN before and after plasma treatments. Before the plasma activation, the contact angle of NT-Cu and SiCN were respectively 54.8° and 51.1° but significantly decreased after the plasma treatments.

SiCN/SiCN before and after post annealing at 250 °C for 1 h. As shown in Fig. 13(a), some nanovoids were observed at the bonding interface after room temperature bonding. At this point, these voids formed due to the limited contact area between the two surfaces. However, these nanovoids were eliminated after annealing at 250 °C for 1 h, as shown in Fig. 13(b). This was because the SiCN contained numerous dangling

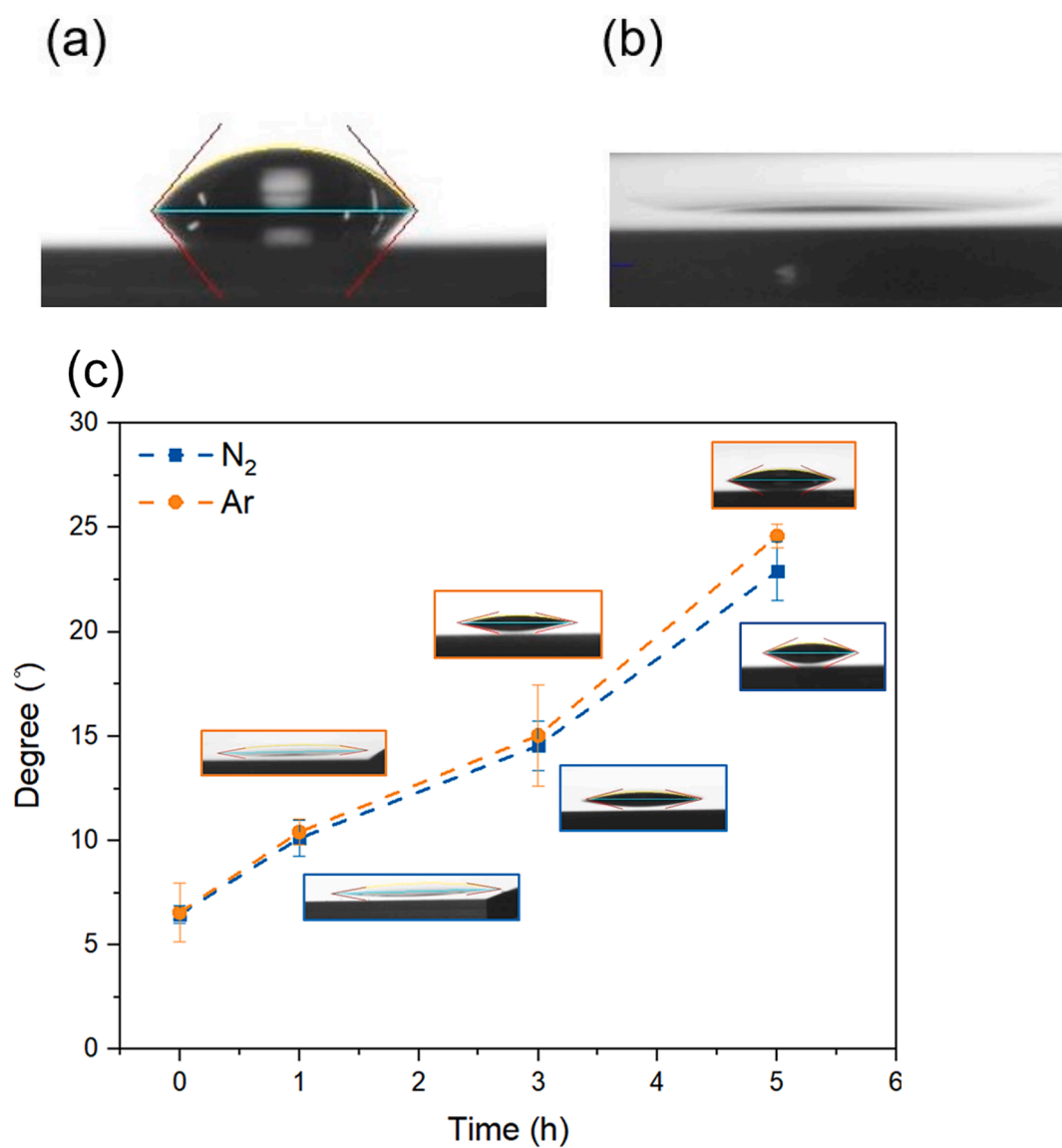


Fig. 8. Contact angles of the SiCN (a) before and (b) after plasma treatments. (c) shows the changes in the SiCN contact angles with storage time after N₂ and Ar plasma treatments at 20 W.

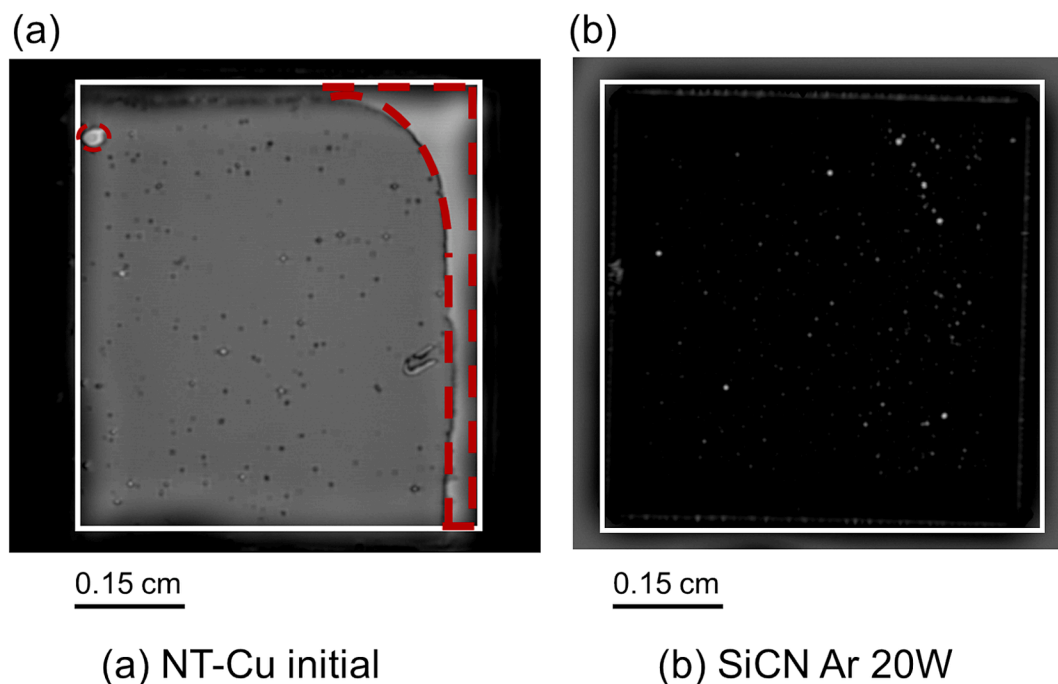


Fig. 9. SAT image of (a) NT-Cu without plasma treatment bonded at 200 °C for 30 min under 33 MPa and (b) SiCN treated by Ar 20 W and bonded at 150 °C for 30 min under 33 MPa. The red dashed line denotes the poorly bonded area.

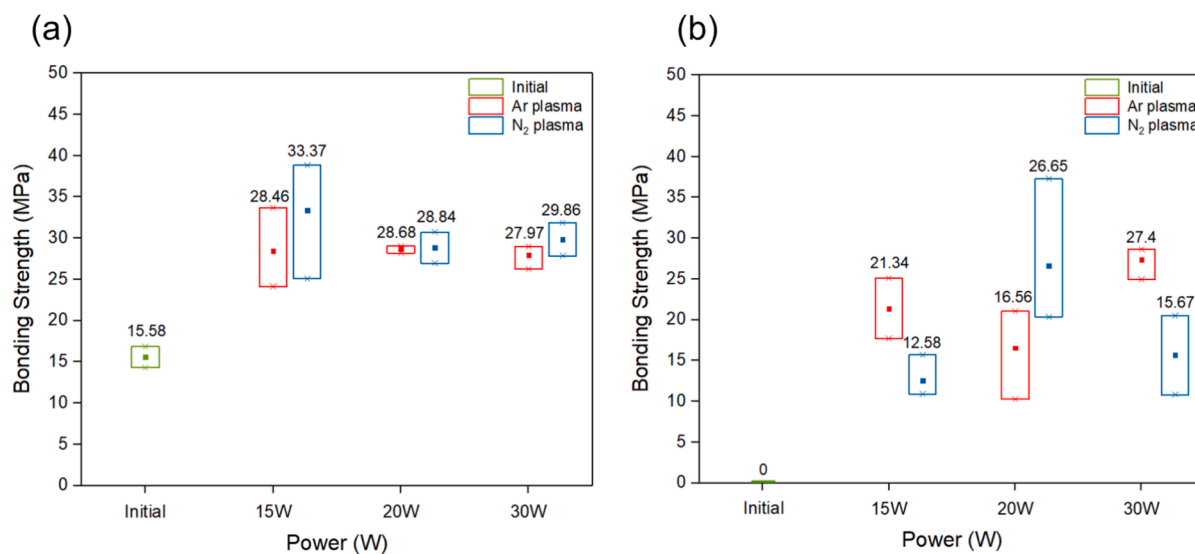


Fig. 10. The bonding strength of the (a) NT-Cu and (b) SiCN joints treated by different plasma powers. The bonding strength of the initial NT-Cu was 15.58 MPa, increasing to 30 MPa after the plasma activation. The SiCN joints without plasma treatment failed during SAT analysis, resulting in no measurable bonding strength. Nevertheless, after plasma treatment, the bonding strength of 27.4 MPa was achieved.

bonds. These dangling bonds in the SiCN films could react with excess water to form Si-OH bonds. The water could originate from those trapped at the bonding interface or from the polymerization of two silanol bonds. However, this process has an energy barrier of 0.8 eV [8,21], making it unlikely to occur at room temperature. During annealing at 250 °C for 1 h, the dangling bonds could gain enough energy to overcome the energy barrier. Thus, the interfacial nanovoids could be eliminated. Fig. 14 shows the EDS analysis of the SiCN/SiCN bonding interfaces. The Si, C, and N contents decreased at the bonding interfaces with the obvious increase in the O content, indicating that an oxide layer formed even at room temperature.

Table 1 shows the comparison of the bonding energy of our SiCN/

SiCN joints with that reported in the previous studies. It can be seen that different testing methods yield varied results. Furthermore, the bonding energy also depends on the annealing temperature and duration. We demonstrate that the bonding energy can reach 1.2 J/m² after annealing at 100 °C for just 30 min. Increasing the annealing temperature or extending the annealing time may result in higher bonding energy. However, the Si substrate may delaminate before the debonding of the SiCN interfaces during four-point bending at high annealing temperatures.

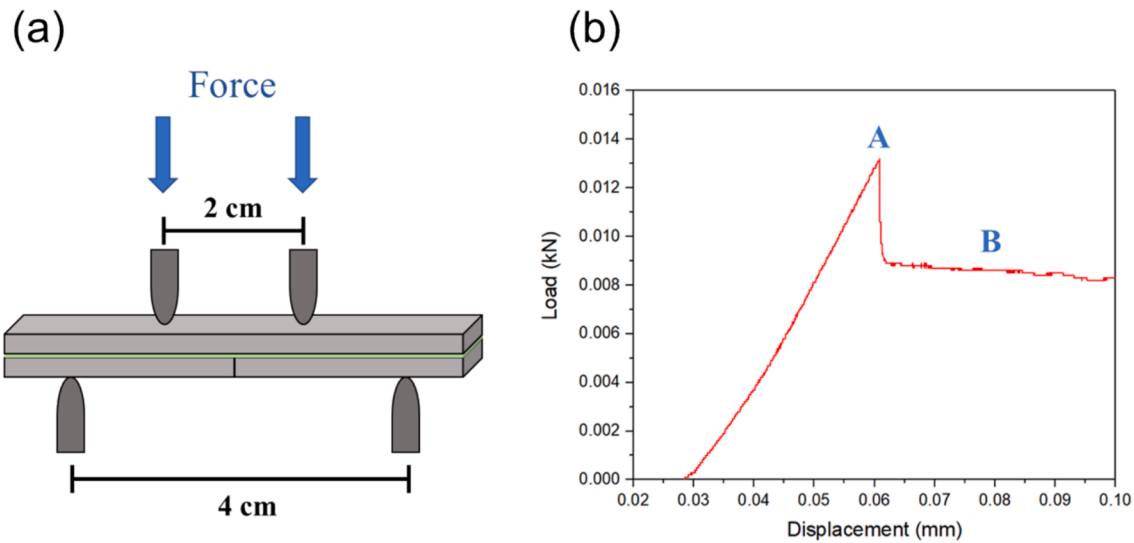


Fig. 11. (a) Schematic illustration of the four-point bending measurements. A notch was firstly cut. (b) The load vs. displacement curve of four-point bending. Point A is the moment when the notch thoroughly cracks, and plateau area B is where the interface continues to delaminate.

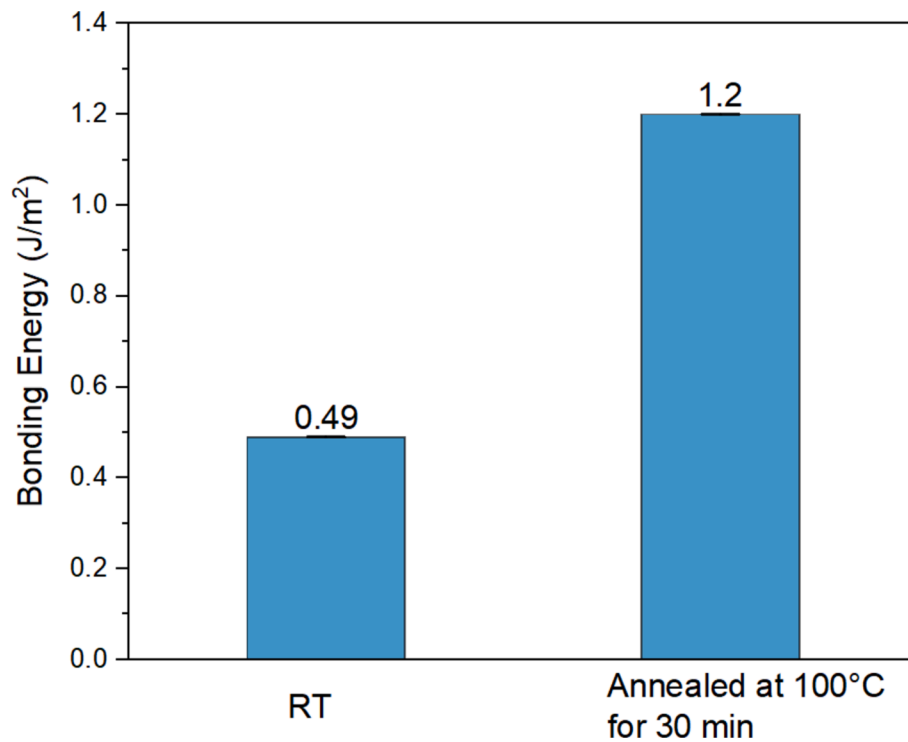


Fig. 12. Bonding energies of the joints bonded at room temperature and after annealing at 100 °C for 30 min. The bonding energy of RT-bonded SiCN was 0.49 J/m²; after annealing at 100 °C for 30 min, it increased to 1.2 J/m².

4. Conclusions

The bonding strength of the NT-Cu/NT-Cu and SiCN/SiCN joints was enhanced using N₂ and Ar plasma activation. The XPS analysis shows that the surface of NT-Cu contained Cu and cuprous oxide. The plasma treatments broke the Si-N and Si-C bonds, leading to the formation of a substantial number of Si-O bonds. The intensity of Si-O bonds increased as the plasma power increased. The bonding strength of SiCN/SiCN was thus enhanced up to 27.4 MPa, which could be attributed to the formation of Si-O bonds. AFM analysis and contact angle measurements showed that the plasma activation process did not obviously damage the surfaces. After the plasma treatments, the roughness of NT-Cu was still

below 4 nm, whereas that of SiCN remained below 0.5 nm. The contact angles of the NT-Cu films significantly decreased from 54.8° to 32.1°, and that of the SiCN decreased from 51.1° to below 10°. In addition, we achieved the room temperature bonding of SiCN/SiCN using the Ar and N₂ plasma. Four-point bending tests were conducted to measure the bonding energies. It was 0.49 J/m² at room temperature and increased to 1.2 J/m² after additional annealing at 100 °C for 30 min. Some nanovoids were found in the joints after room temperature bonding. Those voids could be eliminated by annealing at 250 °C for 1 h.

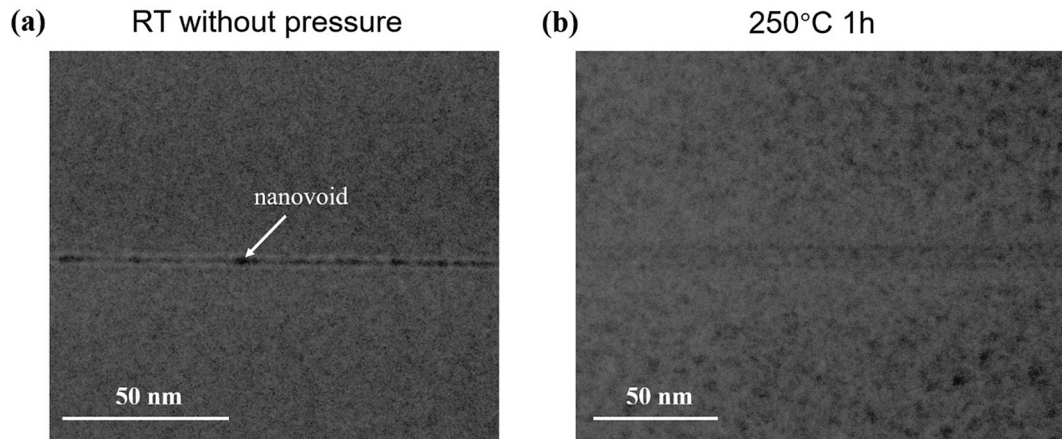


Fig. 13. TEM dark field image of the SiCN treated by Ar plasma 20 W for 1 min and N₂ plasma 20 W for 2 min and bonded in (a) room temperature and (b) post-annealing at 250 °C for 1 h. After RT bonding, obvious nanovoids were found at the bonding interface. Those nanovoids could be eliminated by annealing at 250 °C for 1 h.

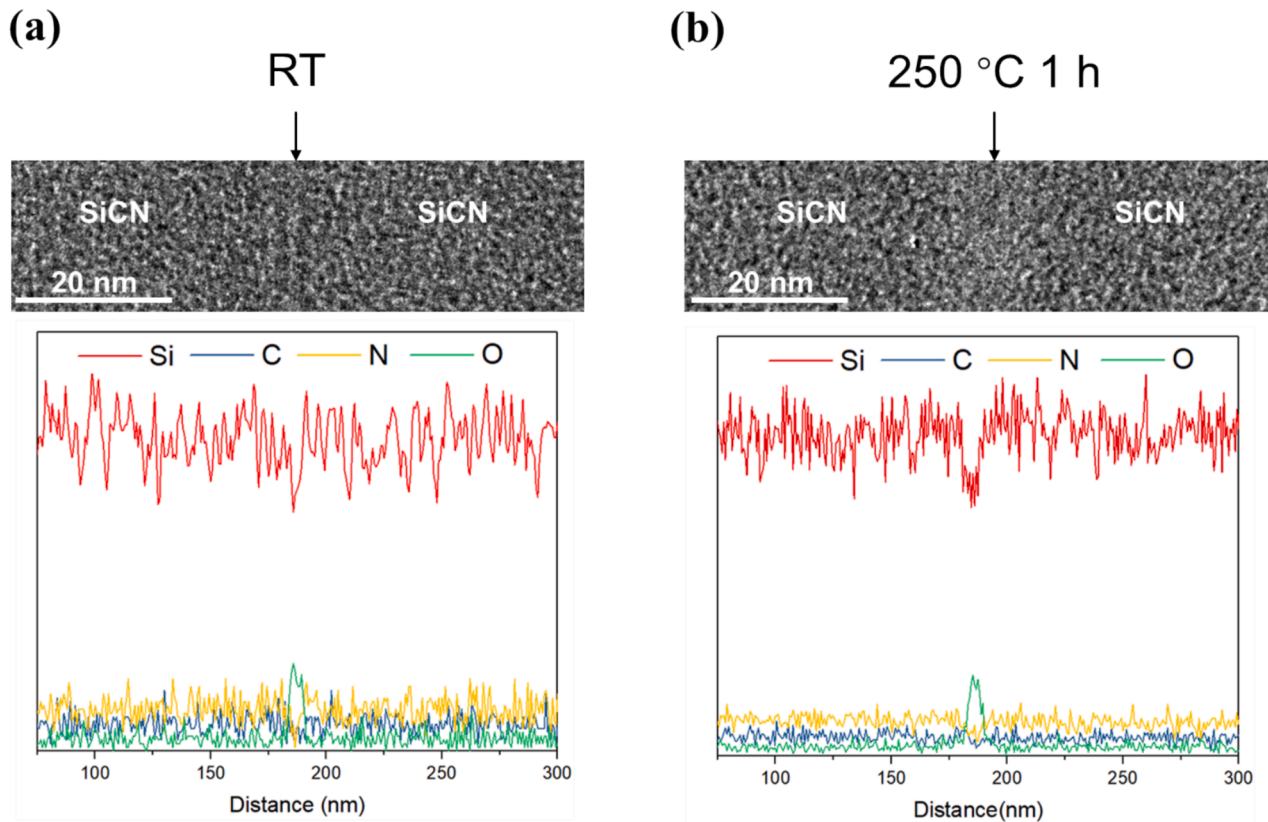


Fig. 14. EDS analyses of the SiCN treated by Ar plasma 20 W for 1 min and N₂ plasma 20 W for 2 min and bonded in (a) room temperature and (b) post-annealing at 250 °C for 1 h. At both interfaces, oxygen signals were obviously detected, suggesting that the bonding interfaces consisted of an oxide layer.

Table 1

Comparison of the bonding energy of SiCN joints in the recent studies.

Materials	Testing Method	Annealing Temp. (°C)/Time (h)	Bonding Energy (J/m ²)	Ref.
SiCN/SiCN	4 PB		1.75–2.25	
	DCB	125 °C–200 °C/2 h	~ 3	[37]
	DCB	420 °C/2 h	2.99	[38]
	DCB	350 °C/1 h	~ 0.5–0.8	[14]
	DCB	350 °C	0.9–1.5	[28]
	modified DCB	NA	~ 2	[39]
	4 PB	100 °C/30 min	1.2	This work

CRediT authorship contribution statement

Rou-Jun Lee: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pin-Syuan He:** Investigation, Formal analysis. **Dinh-Phuc Tran:** Writing – review & editing, Writing – original draft, Validation, Formal analysis, Data curation. **Wei-Lan Chiu:** Resources. **Hsiang-Hung Chang:** Resources. **Chang-Chun Lee:** Resources, Formal analysis. **Chih Chen:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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